

Scale or Shape? Decomposing NTK Movement During Grokking

Group G2

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Abstract

FiXme Fatal: Abstract, written last from the finished sections.

1 Introduction

Grokking is the delayed generalisation that Power et al. (2022) found on small algorithmic datasets: validation accuracy rises from chance long after the training data have been fitted. Modular addition learned by a two-layer perceptron is a minimal instance. Full-batch gradient descent on the squared error groks there with no explicit regularisation (Gromov, 2023), and Kumar et al. (2024) use this setting, with no weight decay, as a baseline.

One account holds that grokking is a transition from lazy training, in which the network behaves as a kernel method with its initial tangent kernel, to rich training, in which the kernel adapts to the task (Kumar et al., 2024). The evidence that the kernel changes is a distance: Mohamadi, Li, et al. (2024) observe that the empirical NTK stays close to its initial value until the training data are fitted and moves far from it afterwards. A distance does not say how the kernel changed. Shan and Bordelon (2022) find that the tangent kernel grows in Frobenius norm during training, as Baratin et al. (2021) report, but that kernel gradient descent with the initial kernel scaled up to the amplitude of the final one still trains more slowly than the network, so that changes to the structure of the kernel, and not its amplitude alone, account for the faster training.

The distinction matters for accounts of what sets the grokking time. The result cited for weight decay forcing the NTK to change (Kumar et al., 2024) is that of Lewkowycz and Gur-Ari (2021), in which the kernel only rescales, and recent accounts time grokking by the weight norm (Khanh et al., 2026), the scale of the logits (Khanh, 2026) and the rate of weight decay (Kim, 2026). None of them measures the shape of the kernel.

This report separates the two with published measures of the tangent kernel (section 3.3): its scale S_t , the Frobenius norm relative to initialisation; its shape change R_t , the kernel distance of Fort et al. (2020), which compares kernels in a scale-invariant manner; its relative variation D_t of Geiger et al. (2020); and its alignment A_t with the labels. We measure these statistics and the kernel’s spectrum along training on the modular-addition baseline of Kumar et al. (2024), and ask which of scale and shape moves before and while the network groks, and whether the times at which it memorises and groks follow either. **FiXme Fatal: State the findings and the contributions once the grid has run.**

2 Related Work

Grokking. Power et al. (2022) report that on small algorithmic datasets, long after a network has overfitted its training data, validation accuracy can rise from chance towards perfect generalisation, and that weight decay is particularly effective at improving generalisation there. Nanda et al. (2023) reverse-engineer the algorithm a small transformer learns for modular addition, which turns addition into rotation about a circle through discrete Fourier transforms, and use it to split training into memorisation, circuit formation and clean-up. For the two-layer perceptron we study, Gromov (2023) gives closed-form weights that solve modular addition when the activation is quadratic, and observes grokking under full-batch gradient descent with no explicit regularisation. Liu, Michaud, and Tegmark (2023) trace grokking to a mismatch between the training and test losses as functions of the weight norm.

Lazy and rich training. In the infinite-width limit the NTK stays constant during training, so the network trains as a kernel method (Jacot, Gabriel, and Hongler, 2020). Chizat, Oyallon, and Bach (2020) show that this lazy regime is a matter of scale: multiplying a model by a factor α , with its initial output kept bounded, brings its training closer to that of its linearisation as α grows, while the mean-field factor $1/m$ of a two-layer network of width m keeps the dynamics non-degenerate as $m \rightarrow \infty$. Kumar et al. (2024) propose that grokking is a transition from lazy to rich, feature-learning dynamics, which arises when the top eigenvectors of the initial NTK are misaligned with the labels, the dataset is large enough to generalise from but not so large that the training loss tracks the test loss, and training starts in the lazy regime. Mohamadi, Li, et al. (2024) observe that the Frobenius distance of the empirical NTK from its initial value is negligible until the training data are fitted and large afterwards, and prove that with fewer than Cp^2 training pairs, for a constant C they do not give, no input-output permutation-equivariant kernel method has expected loss below half that of the zero predictor on modular addition. Lyu et al. (2024) prove that homogeneous networks trained with large initialisation and small weight decay stay at a kernel predictor for a long time and then move sharply to a min-norm or max-margin predictor. Atanasov, Bordelon, and Pehlevan (2021) describe silent alignment, in which the kernel’s eigenstructure evolves while the kernel is small and only its overall scale grows afterwards, under small initialisation and whitened data.

Kernel alignment and spectral bias. On a fixed kernel, gradient descent fits the components of the labels along the eigenvectors with large eigenvalues first (section 3.4). Networks learn low frequencies first (Rahaman et al., 2019), which Basri et al. (2019) explain for a two-layer network on the sphere through the eigenvalues of its NTK at each frequency, and Canatar, Bordelon, and Pehlevan (2021) relate the generalisation of kernel regression to the share of the target’s power in the top eigenfunctions. Outside the kernel regime, the top eigenfunctions of the NTK align with the target during training (Kopitkov and Indelman, 2020), the NTK changes rapidly in a short initial transient and at constant velocity afterwards (Fort et al., 2020), and the spectrum of the tangent kernel becomes more anisotropic while its centred alignment with the class labels rises (Baratin et al., 2021). Shan and Bordelon (2022) give a theory of this alignment for deep linear and two-layer ReLU networks, in which, with several outputs, the kernel of each output aligns with its own target. Kumar et al. (2024) also compute an alignment of the initial kernel with the labels, on the test set; their printed formula has no centring, and they evaluate it at initialisation only.

What sets the grokking time. Khanh et al. (2026) clamp the weight norm at a multiple ρ of a critical norm and find that the time to grok grows exponentially in ρ . Khanh (2026) show that under cross-entropy the norm acts on this delay through the scale of the logits, while under squared error its effect is about half as large and runs through a different route. Kim (2026) derive, for linear models trained with heavy-ball momentum β and weight decay, a grokking time that scales as $(1 - \beta)/(\eta\lambda)$, and observe this scaling on modular addition. Lewkowycz and Gur-Ari (2021) prove, assuming a conjecture on the correlation functions of wide networks, that for a k -homogeneous network in the infinite-width limit trained by gradient flow, L_2 regularisation makes the NTK evolve as $\frac{d}{dt}\Theta_t = -2(k - 1)\lambda\Theta_t$, and that under squared error its eigenvectors stay fixed while its eigenvalues decay; Kumar et al. (2024) cite this result, in an argument they call speculative, for weight decay moving networks out of the lazy regime. The evolution in that result is a rescaling of the kernel. Its proof starts from the gradient-flow equation of the kernel at finite width,

$$\frac{d\Theta_t(x, x')}{dt} = -2(k - 1)\lambda\Theta_t(x, x') + T_t(x, x') + T_t(x', x), \quad (1)$$

where $T_t(x, x') = -\sum_{x''} \partial_{\mu\nu} f_t(x) \partial_{\mu} f_t(x') \partial_{\nu} f_t(x'') \ell'(x'')$ sums over the training inputs and repeated parameter indices, and shows that T_0 has expectation of order n^{-1} in the width n , so that T drops out in the limit (Lewkowycz and Gur-Ari, 2021). In (1) the weight decay appears explicitly only in the term that rescales the kernel; T_t depends on it only through the trajectory. For homogeneous networks trained on the squared error with weight decay, Pracher et al. (2026) show that a residual persists after the training data are fitted, write the evolution of the kernel as a term driven by that residual plus a weight-decay term that shrinks the kernel, and predict a grokking time inversely proportional to the product of learning rate and weight decay, which they observe on modular addition; Xu, Vardi, and Safran (2026) prove bounds on the grokking time of over-parameterised linear regression under gradient descent with weight decay. Both mechanisms need weight decay, and none of these accounts separates the scale of the kernel from its shape.

3 Methods

3.1 Task, network and training

The task is the modular addition of Gromov (2023) at the baseline of Kumar et al. (2024). For a modulus p , each pair (a, b) of residues in $\{0, \dots, p-1\}$ is encoded as the concatenation $x_{a,b} \in \mathbb{R}^{2p}$ of the one-hot codes of a and b , and its target $y_{a,b} \in \mathbb{R}^p$ is the one-hot code of $(a+b) \bmod p$. The dataset holds all p^2 pairs. Each seed draws its own split of the pairs into a training set $\mathcal{T}$ and a test set, as well as its own initial weights.

The network is the two-layer perceptron without biases of Gromov (2023),

$$f(x; \theta) = c_N W^{(2)} \phi(D^{-1/2} W^{(1)} x), \quad (2)$$

with input dimension $D = 2p$, width N , weights $W^{(1)} \in \mathbb{R}^{N \times D}$ and $W^{(2)} \in \mathbb{R}^{p \times N}$ whose entries are drawn from $\mathcal{N}(0, 1)$, and an activation ϕ applied to each entry. Gromov (2023) derives his exact solution for the quadratic activation, finds his empirical results only mildly sensitive to the choice of activation, and reports that the same weights work for the rectifier at a larger width; we use the rectifier $\phi(h) = \max\{h, 0\}$. A network function is k -homogeneous if scaling every parameter by $c > 0$ scales it by c^k , and a fully connected network with L layers and rectifier activations is L -homogeneous (Lewkowycz and Gur-Ari, 2021), so (2) is 2-homogeneous; the centred predictor below, which subtracts the constant $f(x; \theta_0)$, is not. The readout factor is $c_N = N^{-1/2}$ in the NTK parameterisation of Jacot, Gabriel, and Hongler (2020) and $c_N = N^{-1}$ in the mean-field parameterisation of Gromov (2023). Following Kumar et al. (2024), the trained predictor is the centred and rescaled function

$$\tilde{f}(x; \theta) = \alpha [f(x; \theta) - f(x; \theta_0)], \quad (3)$$

where θ_0 holds the initial weights, and the learning rate is $\eta = \eta_0/\alpha^2$, which keeps the initial rate of change of the function the same for every α . The loss is the squared error averaged over the p outputs and the training pairs, $L(\theta) = \frac{1}{p|\mathcal{T}|} \sum_{(a,b) \in \mathcal{T}} \|\tilde{f}(x_{a,b}; \theta) - y_{a,b}\|^2$, and training is full-batch gradient descent with weight decay,

$$\theta_{k+1} = (1 - \eta\lambda) \theta_k - \eta \nabla L(\theta_k). \quad (4)$$

This is the decoupled weight decay of Kim (2026) with no momentum buffer, and for plain gradient descent coupled L_2 regularisation and decoupled weight decay coincide (Kim, 2026). The grid sets the product $\eta\lambda$ directly.

3.2 The empirical tangent kernel

For a scalar output $g(x; \theta)$ the tangent features are the gradients $\nabla_{\theta} g(x; \theta)$, and the tangent kernel is the inner product of the features of two inputs (Baratin et al., 2021; Jacot, Gabriel, and Hongler, 2020). We evaluate it on all $n = p^2$ pairs, so that at step t the kernel is the matrix $K_t \in \mathbb{R}^{n \times n}$ with entries

$$[K_t]_{ij} = \langle \nabla_{\theta} g(x_i; \theta_t), \nabla_{\theta} g(x_j; \theta_t) \rangle. \quad (5)$$

Two scalar outputs are measured: the sum of the logits of (2) divided by $\sqrt{p}$, whose kernel is the pseudo-NTK of Mohamadi, Bae, and Sutherland (2022) and is reported in the main text, and the first logit, which Mohamadi, Li, et al. (2024) use in place of the full kernel of a vector output. The kernel is evaluated at initialisation, on a logarithmically spaced grid of steps, and at every event step of section 3.5.

3.3 Scale, shape and alignment

For a set $\mathcal{S}$ of pairs, $K_t[\mathcal{S}]$ is the submatrix of K_t on those pairs and $Y[\mathcal{S}]$ the matrix whose rows are their targets; $\mathcal{T}$ holds the training pairs of a seed and $\mathcal{V}$ its test pairs. Following Cortes, Mohri, and Rostamizadeh (2024), the Frobenius product is $\langle A, B \rangle_F = \text{tr}(A^\top B)$ with norm $\|A\|_F = \sqrt{\langle A, A \rangle_F}$, and $K_c = CKC$ centres an $r \times r$ kernel matrix with $C = I_r - \frac{1}{r} \mathbf{1}\mathbf{1}^\top$. All four statistics are measured on the kernel of the test pairs of a seed, $K_t = K_t[\mathcal{V}]$ and $Y = Y[\mathcal{V}]$ in this subsection. We record

$$\begin{aligned} S_t &= \frac{\|K_t\|_F}{\|K_0\|_F}, & R_t &= 1 - \frac{\text{tr}(K_t K_0^\top)}{\sqrt{\text{tr}(K_t K_t^\top)} \sqrt{\text{tr}(K_0 K_0^\top)}}, \\ D_t &= \frac{\|K_t - K_0\|_F}{\|K_0\|_F}, & A_t &= \text{CKA}(K_t, YY^\top), & \text{CKA}(K, K') &= \frac{\text{tr}(K_c K'_c)}{\|K_c\|_F \|K'_c\|_F}. \end{aligned} \quad (6)$$

The scale S_t is the Frobenius norm of the kernel relative to its value at initialisation. Shan and Bordelon (2022) find that this norm grows during training, and in the silent alignment of Atanasov, Bordelon, and Pehlevan (2021) the kernel first aligns while small and then evolves in scale only, $K(x, x', t) = g(t)K_\infty(x, x')$. The shape change R_t is the kernel distance of Fort et al. (2020) between the kernels at step t and at initialisation, which compares two kernel matrices in a scale-invariant manner. The variation D_t is the total relative variation of the kernel that Geiger et al. (2020) measure on the test set to quantify feature training. The alignment A_t is the centred kernel alignment (CKA) of the kernel with the class label kernel $YY^\top$ on a held-out set, as Baratin et al. (2021) measure it after Cortes, Mohri, and Rostamizadeh (2024); its normalisation makes it invariant under isotropic rescaling (Baratin et al., 2021).

Because the first three statistics are taken on one matrix, the variation is a function of the scale and the shape,

$$D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1, \quad (7)$$

and $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$ (section A.1). A kernel that only rescales therefore has $D_t = |S_t - 1|$, which is as large as its change of scale, so a large D_t does not by itself show that the kernel changed shape.

3.4 Spectral measures

In the linear dynamics of Arora et al. (2019), the predictions u_k on the training pairs follow $u_{k+1} = u_k - \eta G(u_k - y)$ with a fixed Gram matrix G , from $u_0 = 0$. If G has eigenvalues ω_i^G with orthonormal eigenvectors g_i , the residual after k steps is $\|y - u_k\| = \sqrt{\sum_i (1 - \eta \omega_i^G)^{2k} (g_i^\top y)^2}$, so

the part of the labels along an eigenvector with a large eigenvalue is fitted first; Jacot, Gabriel, and Hongler (2020) give the continuous-time form, in which the component along each kernel principal component decays exponentially at a rate set by its eigenvalue. Eigenvalues are written ω to keep them apart from the weight decay λ . The measures use the kernel on all $n = p^2$ pairs, which covers every input of the task. The centred kernel CK_tC , with $C = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, is symmetric and positive semi-definite (Cortes, Mohri, and Rostamizadeh, 2024), so it has orthonormal eigenvectors $v_1, \dots, v_n$ with eigenvalues $\omega_1 \geq \dots \geq \omega_n \geq 0$ (Axler, 2024). We record the target power along each eigenvector, $P_{t,i} = \sum_{c=1}^p (v_i^\top C y_c)^2$, where y_c is the c th column of Y , and its cumulative share over the leading eigenvectors, which Canatar, Bordelon, and Pehlevan (2021) call the cumulative power distribution and use to explain why a kernel generalises on one task and not on another. The centring follows Baratin et al. (2021), who report results for centred tangent features; Jacot, Gabriel, and Hongler (2020) note that their kernel principal component analysis is not centred, so its first component is almost the constant function. From the singular values $\sigma_1 \geq \dots \geq \sigma_n \geq 0$ of CK_tC we compute the effective rank of Roy and Vetterli (2007), which Baratin et al. (2021) apply to the tangent kernel, and from the eigenvalues the trace ratios of Baratin et al. (2021),

$$\text{erank}_t = \exp\left(-\sum_j \mu_j \log \mu_j\right), \quad \mu_j = \frac{\sigma_j}{\sum_i \sigma_i}, \quad T_{t,k} = \frac{\sum_{j \leq k} \omega_j}{\sum_j \omega_j}, \quad (8)$$

which measure how evenly the kernel spreads over its eigenvectors; $T_{t,k}$ is the share of the trace carried by the k largest eigenvalues, which is how Baratin et al. (2021) describe their trace ratios: the relative importance of the top k eigenvalues. Roy and Vetterli (2007) take $0 \log 0 = 0$, so zero singular values, such as the one along the constant vector, which C maps to zero, do not enter the effective rank.

3.5 Event times

Following Khanh et al. (2026), the memorisation step of a seed is the first step at which its training accuracy reaches 0.99, and its grokking step at level ℓ is the first step at which its test accuracy reaches ℓ , for each $\ell \in \{0.80, 0.90 \text{ and } 0.95\}$, the test-accuracy thresholds at which Khanh et al. (2026) check that their critical weight norm does not depend on the threshold. A seed that has not reached a level within the step budget is kept, and its event time is recorded as censored at the budget.

Within a cell, the seeds give the Kaplan–Meier estimate of the probability that an event has not occurred by step t ,

$$\hat{F}(t) = \prod_{j|t_j \leq t} \frac{n_j - d_j}{n_j}, \quad (9)$$

where $t_1 < t_2 < \dots$ are the distinct steps at which the event occurs, d_j seeds reach it at t_j and n_j are still at risk there. Greenwood’s formula estimates its variance as $\hat{F}^2(t) \sum_{j|t_j \leq t} d_j / [n_j(n_j - d_j)]$, and with $\hat{s}^2(t) = \sum_{j|t_j \leq t} d_j / [n_j(n_j - d_j)] / [\log \hat{F}(t)]^2$ the approximate 95 % interval $\hat{F}(t)^{\exp[\pm 1.96\hat{s}(t)]}$ stays within $[0, 1]$ (Kalbfleisch and Prentice, 2002). The median event time of a cell is the step at which $\hat{F}$ falls to one half. A cell whose $\hat{F}$ stays above one half up to the budget has no median within it, and at a grokking level it does not grok.

Each run is also labelled by its phase at the last step, following Pracher et al. (2026): grokking when its training and test accuracies both end above a threshold, memorisation when only its training accuracy does, forgetting when its training accuracy rose above the threshold earlier and ends below it, and no fitting otherwise. The phases describe the cells and do not select runs. A run keeps its event times whatever its phase, since in the analysis of one failure mode a unit

that fails by another mode, or is removed from test without failing, contributes its running time as a censoring time (Nelson, 1990).

Three censored regressions, fitted with lifelines (Davidson-Pilon, 2019), relate the event times to the settings of the cells. The log-normal accelerated failure time model takes the survival function of an event time T with covariates x to be

$$S(t | x) = 1 - \Phi\left(\frac{\log t - \mu(x)}{\sigma}\right), \quad \mu(x) = a_0 + \sum_i a_i x_i, \quad (10)$$

where Φ is the standard normal distribution function, so a unit increase in x_i multiplies the median event time by e^{a_i} . Its parameters are maximum likelihood estimates. With the step budget τ as the common censoring time, seed i contributes the log likelihood

$$\mathcal{L}_i = I_i \left[-\log \sigma - \frac{1}{2} \log(2\pi) - \frac{1}{2} z_i^2 \right] + (1 - I_i) \log[1 - \Phi(\zeta_i)] \quad (11)$$

in terms of its log time, where $z_i = (\log t_i - \mu(x_i))/\sigma$, $\zeta_i = (\log \tau - \mu(x_i))/\sigma$, and $I_i = 1$ if the event is observed and $I_i = 0$ otherwise (Nelson, 1990). This is the likelihood of Kalbfleisch and Prentice (2002) for a log-linear model under independent censoring, with normal errors. The additive model of Aalen (1989) takes the intensity of the event for seed i at step t to be $\alpha_0(t) + \sum_j \alpha_j(t) Z_j^i$ while the seed has not reached the level and zero afterwards, and estimates the cumulative regression functions $A_j(t) = \int_0^t \alpha_j(s) ds$ by

$$A^*(t) = \sum_{T_k \leq t} [Y(T_k)^\top Y(T_k)]^{-1} Y(T_k)^\top I_k, \quad (12)$$

where the event times $T_1 < T_2 < \dots$ are ordered, row i of $Y(t)$ is $(1, Z_1^i, \dots, Z_r^i)$ while seed i is at risk and zero otherwise, and I_k is the indicator of the seed whose event occurs at T_k . Unlike (10) with logarithmic covariates, (12) admits a covariate at zero, such as $\eta\lambda = 0$. Intervals come from resampling seeds with the bias-corrected and accelerated bootstrap (Efron and Tibshirani, 1994).

Each event and level is fitted separately. The covariates of (10) and (12) are the settings of a cell, fixed in time. As $\log(\eta\lambda)$ is undefined at $\eta\lambda = 0$, (10) is fitted twice: to the cells without weight decay, with indicators of the levels of α , and to the cells with $\eta\lambda > 0$, all at the α of the pilot cell, with the single covariate $\log(\eta\lambda)$, while (12) takes all cells together, with the indicators of α and $\eta\lambda$ itself. Aalen (1989) states his results for covariates fixed in time, and lifelines fits (12) only for such covariates. The statistics of section 3.3 change along training, so they enter a third model, the relative risk model with time-dependent covariates of Kalbfleisch and Prentice (2002),

$$h(t | Z(t)) = h_0(t) \exp[Z(t)^\top \gamma], \quad (13)$$

whose coefficients maximise the partial likelihood

$$L(\gamma) = \prod_{j=1}^k \frac{\exp[Z_j(t_j)^\top \gamma]}{\sum_{l=1}^n Y_l(t_j) \exp[Z_l(t_j)^\top \gamma]}, \quad (14)$$

where $t_1 < \dots < t_k$ are the event times, Z_j belongs to the seed whose event occurs at t_j , and $Y_l(t)$ is one while seed l is at risk. Kalbfleisch and Prentice (2002) write it for untied event times; seeds can reach an event at the same step, and the lifelines fitter handles such ties by Efron's method. Here $Z(t) = (S_t, R_t, A_t)$ is held from each checkpoint to the next, so its paths are left-continuous as the model requires. These covariates are internal: the run that reaches the event generates them. The model therefore describes the rate of the event given the current kernel and has no survivor function of its own. When a setting acts on the event through these covariates, an analysis conditional on them shows little effect of that setting (Kalbfleisch and Prentice, 2002).

3.6 Choice of levels

Where weight decay sets the grokking time, (10) with the single covariate $x = \log(\eta\lambda)$ makes $\log T$ normal with mean $a_0 + a_1x$ and variance σ^2 . Without censoring this is the simple linear regression model, whose designs Berger and Wong (2009) construct as follows. A design ξ places a proportion w_j of the runs at level x_j , $j = 1, \dots, m$, with $\sum_j w_j = 1$, on levels rescaled to $[-1, 1]$, and has the information matrix

$$M(\xi) = \sum_{j=1}^m w_j f(x_j) f(x_j)^\top, \quad f(x) = (1, x)^\top. \quad (15)$$

With N runs, $n_j = Nw_j$ of them at x_j , the least squares estimates have covariance $\sigma^2(X^\top X)^{-1}$, where $X^\top X = NM(\xi)$, and the slope has variance $\text{var}(\hat{a}_1) = \sigma^2 / \sum_i (x_i - \bar{x})^2$. The design that minimises this variance is unique and places half the runs at each end of the interval, and for the straight line it is also D-, A-, E- and G-optimal. When the line is in doubt, a quadratic term measures its curvature. The D-optimal design for the quadratic places a third of the runs at each of -1 , 0 and 1 , and the design for its curvature coefficient alone places half of them at 0 and a quarter at each end. For many models with p parameters an optimal design has between p and $p(p+1)/2$ distinct levels, and a design ξ is compared with the D-optimal design ξ^* of the model that holds by its relative efficiency

$$\text{RE}_D = \left\{ \frac{\det \text{Cov}(\hat{a}^*)}{\det \text{Cov}(\hat{a})} \right\}^{1/p}, \quad (16)$$

where $\hat{a}^*$ and $\hat{a}$ are the estimates under ξ^* and ξ .

Censoring at the budget changes the information a run carries. For $\mu(x) = a_0 + a_1x$ and censoring at $\log \tau$, a run at level x contributes to the parameters (a_0, a_1, σ) the expected information

$$F_x = \frac{1}{\sigma^2} \begin{pmatrix} A(\zeta) & xA(\zeta) & B(\zeta) \\ xA(\zeta) & x^2A(\zeta) & xB(\zeta) \\ B(\zeta) & xB(\zeta) & C(\zeta) \end{pmatrix}, \quad (17)$$

where $\zeta = (\log \tau - \mu(x))/\sigma$ and, with ϕ and Φ evaluated at ζ ,

$$\begin{aligned} A(\zeta) &= \Phi - \left[\zeta - \frac{\phi}{1 - \Phi} \right] \phi, & B(\zeta) &= -\phi \left[1 + \zeta \left(\zeta - \frac{\phi}{1 - \Phi} \right) \right], \\ C(\zeta) &= 2\Phi - \zeta\phi \left[1 + \zeta^2 - \frac{\zeta\phi}{1 - \Phi} \right]. \end{aligned} \quad (18)$$

The information of a plan is the sum of F_x over its runs, the covariance of the maximum likelihood estimates is its inverse, and a function h of the parameters has variance $HF^{-1}H^\top$, where H is the gradient of h (Nelson, 1990). Because ζ depends on the parameters, an optimum plan for censored data is found by numerical search for guessed values of them and is only locally optimum, and good compromise plans run more units at low stress, here small $\eta\lambda$, than at high (Nelson, 1990).

Across the level at which the grokking time returns to its value without weight decay, the mean log time is a broken line, $\mu(x) = a_0 + a_1x + b(x - \psi)_+$ with $(x - \psi)_+ = (x - \psi)I(x > \psi)$. The method of V. M. R. Muggeo (2003) applies to any regression model with a linear predictor, survival models included, so it runs inside the censored likelihood of (10). Step s of the fit sets $U^{(s)} = (x - \psi^{(s)})_+$ and $V^{(s)} = -I(x > \psi^{(s)})$, fits the linear predictor $a_1x + bU^{(s)} + \kappa V^{(s)}$, and updates $\psi^{(s+1)} = \psi^{(s)} + \hat{\kappa}/\hat{b}$ until $\hat{\kappa}$ vanishes, when every estimate, $\hat{\psi}$ included, is maximum likelihood. Fixing the left slope a_1 at zero gives a threshold model, in which the grokking time

does not change below the break, and whether the left slope is zero is a Wald test of a_1 in the unconstrained fit (V. M. R. Muggeo, 2008). Whether a break exists at all is not a Wald test of b , since ψ vanishes under $b = 0$; the segmented package bounds the p -value of that test by the method of Davies (V. M. R. Muggeo, 2008). The break has the Wald interval $\hat{\psi} \pm 1.96 \text{SE}(\hat{\psi})$, whose standard error comes from the delta method for the ratio $\hat{\kappa}/\hat{b}$ and, at convergence, where $\hat{\kappa} = 0$, reduces to $\text{SE}(\hat{\kappa})/|\hat{b}|$ (V. M. R. Muggeo, 2003, 2008). When the break lies away from the middle of the range of x , its estimate is biased with an asymmetric distribution and the Wald interval can perform poorly, in part because $\text{SE}(\hat{\psi})$ may understate its variability (V. M. Muggeo, 2017). The break depends on constants that the grid itself estimates, so a design for it is also only locally optimal; a sequential design takes the guesses from a first stage, which here consists of the pilot runs, and updates the design with their estimates (Berger and Wong, 2009).

Park (1978) constructs D-optimal designs for segmented polynomials with a known break. For k distinct levels, $|X^\top X|$ is largest when every level has the same number of runs, and under continuity at the break the design is D-optimal on each segment with a level at the break. A line below the break and a quadratic above it therefore have the design $x \in \{-1, \psi, (1 + \psi)/2, 1\}$ on the rescaled axis. The quadratic is one degree above the line expected above the break, as Park (1978) advises when lack of fit is to be tested, with equal numbers of runs at each level. The design is locally optimal at the guessed break. The guess is the product $\eta\lambda$ at which the inverse-decay boundary that Pracher et al. (2026) calibrate for a network on the same task, scaled by their clock, gives a grokking time equal to the pilot’s time without weight decay. A second stage places the design again at the estimated break.

4 Experiments

4.1 The pilot cell

The pilot cell is the baseline of Kumar et al. (2024): modulus $p = 23$, a fraction 0.90 of the pairs for training, width $N = 100$ in the NTK parameterisation, $\alpha = 1$, $\eta_0 = 100$, $\eta\lambda = 0$ and a budget of 100 000 steps, for 5 seeds. Table 1 gives the steps at which the seeds memorise and grok, Figure 1 the losses and the statistics of the sum kernel along training, and Table 2 those statistics and the spectral measures at initialisation and at the last step. **FiXme Fatal: Describe the pilot against the questions of the introduction once the grid gives the comparison cells.**

4.2 The grid

FiXme Fatal: The grid: its arms and levels, each level taken from a transcribed source, and the timing fits.

Table 1: Steps at which the seeds of the pilot cell first reach each event: training accuracy at the memorisation level, or test accuracy at a grokking level. Median, minimum and maximum over seeds, and the number of seeds that reach the event within the budget.

Event	Level	Median	Minimum	Maximum	Reached
memorised	0.99	2600	2500	2800	5
grokked	0.80	15 000	14 000	32 000	5
grokked	0.90	18 000	17 000	38 000	5
grokked	0.95	34 000	22 000	42 000	5

Table 2: Statistics (6) and spectral measures (8) of the sum kernel of the pilot cell at initialisation and at the last step, and the share of the centred target’s power on the $2(p-1)$ leading eigenvectors. Minimum, median and maximum over seeds.

Statistic	Step	Minimum	Median	Maximum
A_t	0	0.39	0.40	0.40
A_t	100 000	0.32	0.34	0.36
D_t	0	0	0	0
D_t	100 000	2.9	3.1	3.4
R_t	0	0	0	0
R_t	100 000	0.066	0.068	0.080
S_t	0	1	1	1
S_t	100 000	3.9	4	4.4
erank_t	0	120	120	130
erank_t	100 000	51	53	54
target power on $v_1, \dots, v_{2(p-1)}$	0	0.0016	0.0017	0.0021
target power on $v_1, \dots, v_{2(p-1)}$	100 000	0.019	0.020	0.021
$T_{t,2(p-1)}$	0	0.76	0.77	0.78
$T_{t,2(p-1)}$	100 000	0.92	0.93	0.93

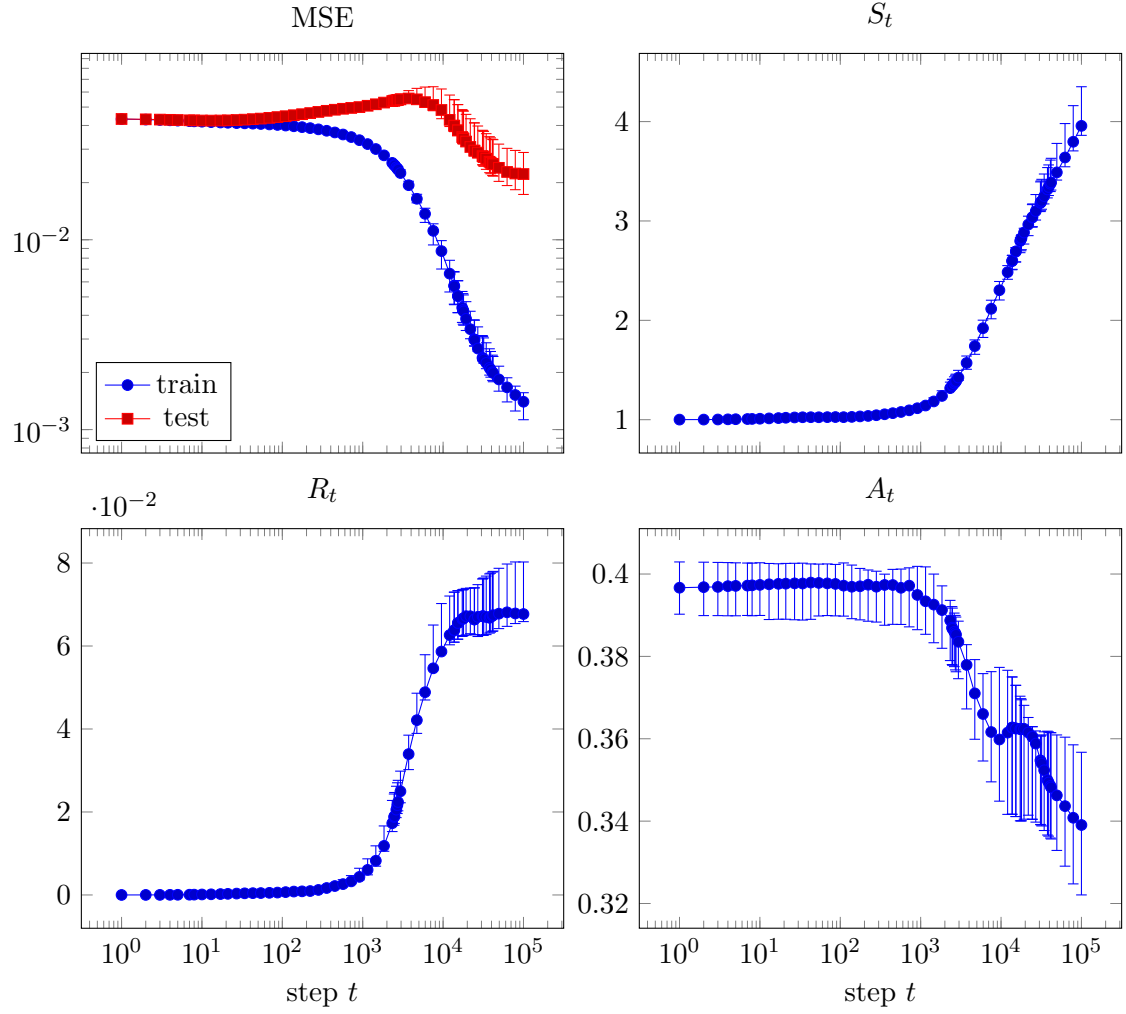


Figure 1: Train and test MSE and the statistics S_t , R_t and A_t of the sum-over-logits kernel against step t , at the baseline cell of Kumar et al. (2024). Points are medians over 5 seeds and error bars span their range.

5 Conclusion

FiXme Fatal: Conclusion: what the measurements show, and the limitations.

Contribution

FiXme Fatal: Contribution of each group member, from the group.

Acknowledgement

FiXme Fatal: Acknowledgement, written by the group.

A Derivations

A.1 Scale, shape and variation

Throughout, K_t and K_0 are real $r \times r$ kernel matrices on the test pairs of a seed with $K_0 \neq 0$ and $K_t \neq 0$, so that (6) is defined.

Results used. On the linear maps between two finite-dimensional inner product spaces, $\langle S, T \rangle = \text{tr}(T^* S)$ is an inner product, which with respect to orthonormal bases is the standard inner product of the matrix entries, and its norm is the Frobenius norm (Axler, 2024). For real matrices the adjoint is the transpose, so this is the Frobenius product $\langle A, B \rangle_F = \text{tr}(A^\top B)$ of Cortes, Mohri, and Rostamizadeh (2024). Square matrices satisfy $\text{tr}(AB) = \text{tr}(BA)$ (Axler, 2024). In an inner product space,

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle,$$

over the real numbers $\langle u, v \rangle = \langle v, u \rangle$, and $\langle u, v \rangle = \|u\| \|v\|$ holds if and only if one of u and v is a nonnegative real multiple of the other (Axler, 2024).

The identity. By $\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(K_t K_0^\top) = \text{tr}(K_0^\top K_t) = \langle K_0, K_t \rangle_F$, and likewise $\text{tr}(K_t K_t^\top) = \|K_t\|_F^2$, so (6) gives

$$1 - R_t = \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \|K_0\|_F}.$$

The expansion above with the symmetry of the real inner product gives

$$\|K_t - K_0\|_F^2 = \|K_t\|_F^2 + \|K_0\|_F^2 - 2\langle K_t, K_0 \rangle_F.$$

Dividing by $\|K_0\|_F^2$, the left side is D_t^2 , the first term is S_t^2 , and

$$\frac{\langle K_t, K_0 \rangle_F}{\|K_0\|_F^2} = \frac{\|K_t\|_F}{\|K_0\|_F} \cdot \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \|K_0\|_F} = S_t(1 - R_t),$$

which gives (7).

When the shape is unchanged. By the expression for $1 - R_t$ above, $R_t = 0$ exactly when $\langle K_t, K_0 \rangle_F = \|K_t\|_F \|K_0\|_F$, that is, by the last result above, exactly when one of K_t and K_0 is a nonnegative real multiple of the other. Neither is zero, so the multiple is positive, and $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$. Then $S_t = c$, and (7) gives $D_t^2 = (S_t - 1)^2$.

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